

PANCREAS (Surgery) — Revision Table

Bold = high-yield. Focus: acute pancreatitis.

ANATOMY & INVESTIGATION

Topic	Key points
Anatomy	Retroperitoneal; head 30%, body+tail 70%; 80–90% exocrine acinar
Islets	75% B (insulin), 20% A (glucagon) , D (somatostatin), PP
Enzymes	Amylase most used (rises hours, falls 4–8d); lipase more sensitive/specific
↑Amylase (non-pancreatitis)	Upper GI perforation, mesenteric infarction, ectopic pregnancy, macroamylasaemia, renal failure, salivary

ACUTE PANCREATITIS — DIAGNOSIS & AETIOLOGY

Topic	Key points
Definition (2 of 3)	Pain + amylase/lipase ≥3× normal + CT findings
Mechanism	Premature enzyme activation → autodigestion → oedema/necrosis
Categories	Mild (interstitial, mortality ~1%); severe (necrotising, 5–10%, mortality 20–50%)
Aetiology	Gallstones 50–70% + alcohol 25%; I GET SMASHED (Idiopathic, Gallstones, Ethanol, Trauma, Steroids, Mumps/malignancy, Autoimmune, Scorpion, Hypercalcaemia/lipidaemia, ERCP, Drugs)
Note	Gallstone cause → cholecystectomy same admission

CLINICAL, SEVERITY & IMAGING

Topic	Key points
Clinical	Pain (epigastric, radiates to back 50%, refractory) + N/V; Grey Turner's (flank)/Cullen's (umbilicus) = haemorrhage ; pleural effusion 10–20%
Severity	Ranson/Glasgow ≥3 at 48h = severe; CRP >150 at 48h ; persisting organ failure; BMI >30. Glasgow = PANCREAS (PaO2, Age>55, Neutrophils, Calcium, Renal urea, Enzymes LDH, Albumin, Sugar)
Atlanta (2013)	Mild (no organ failure); moderate (transient <48h); severe (persistent >48h)
Imaging	Plain film: sentinel loop, colon cut-off, renal halo . US within 24h (gallstones). CECT (Balthazar) if doubt/severe/complications; underestimates necrosis first 72h

MANAGEMENT & COMPLICATIONS

Topic	Key points
Mild	Conservative: IV fluids, NO antibiotics , analgesia/antiemetic, brief fast
Severe	ICU + aggressive fluids + O2 + monitoring ; antibiotics if cholangitis; ERCP <72h (gallstone + cholangitis); enteral (NG) feeding
Systemic complications	Shock, ARDS, renal failure, DIC, hypocalcaemia (first week)
Necrosis	No CT enhancement → ANC → walled-off necrosis (4 wks); infected necrosis = 50% mortality; sterile necrosis NOT drained
Pseudocyst	Amylase-rich, fibrous wall, needs ≥4 weeks ; complications: infection, rupture, haemorrhage (vessel erosion), obstruction
Other local	Abscess, ascites, pleural effusion, portal/splenic vein thrombosis, pseudoaneurysm